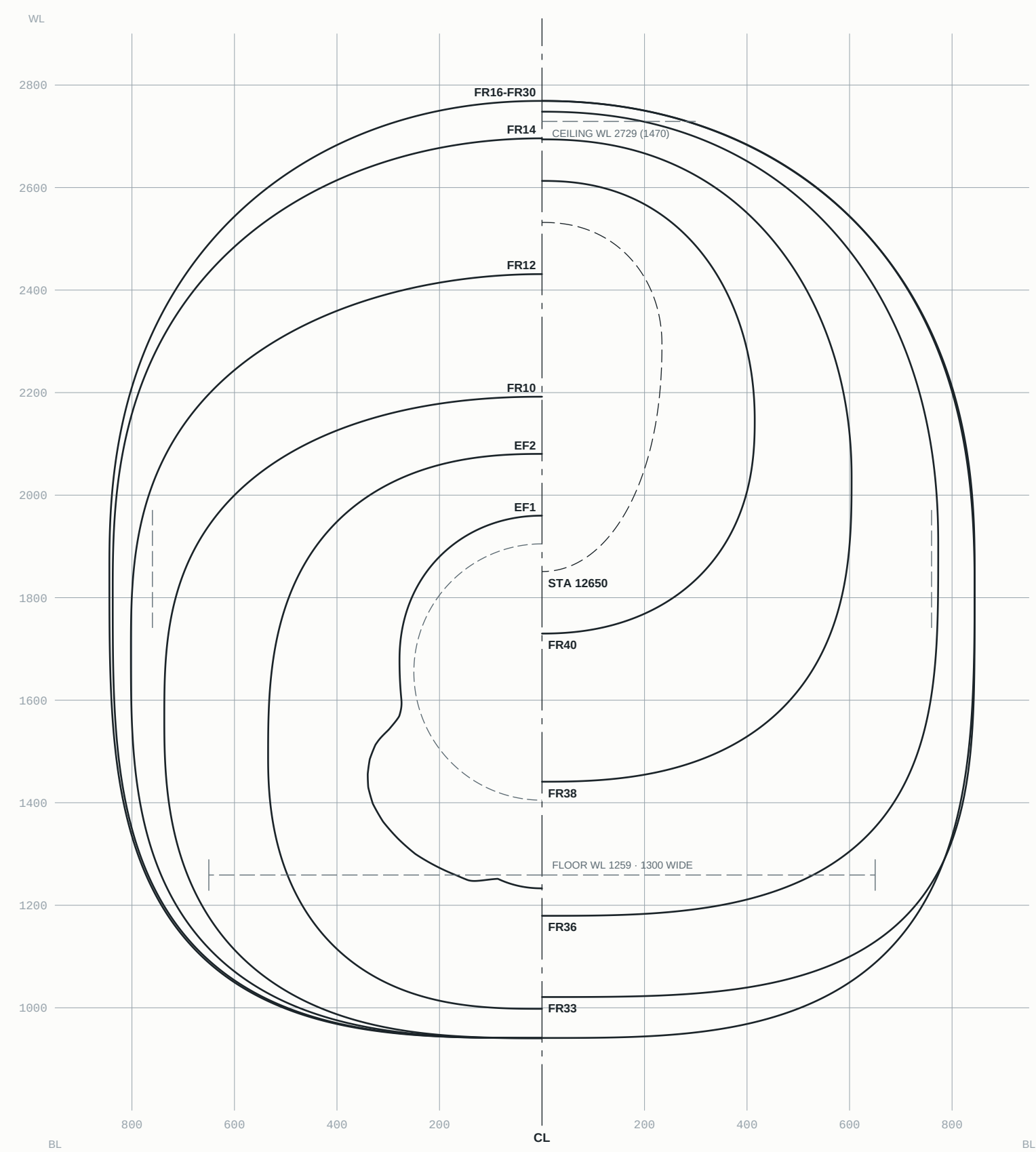
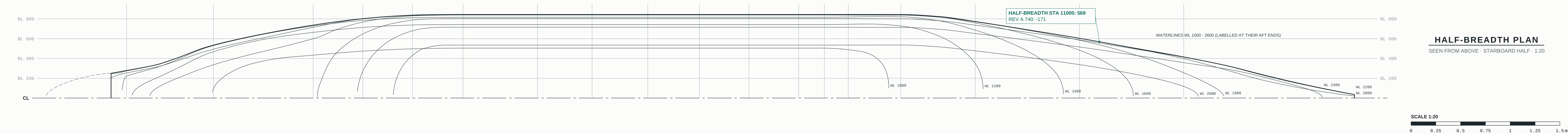
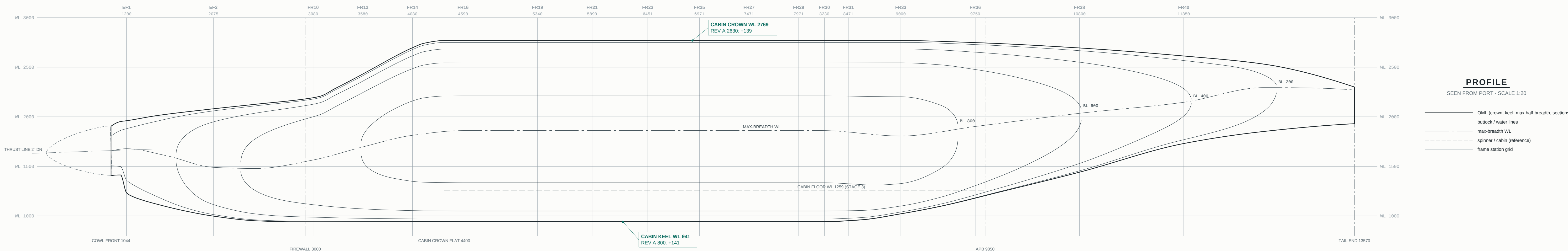




CONTROL-LINE KNOTS (model/fuselage.py)								
STA	CROWN	KEEL	HALF-BR	MAX-BR WL	n TOP	m TOP	n BOT	m BOT
1044	1095 ⁺	1485 ⁺	258 ⁺	1055 ⁺	2.08 ⁺	2.00 ⁺	2.08 ⁺	2.00 ⁺
1140	1202 ⁺	1413 ⁺						
1200	1960	1233	278	1678	2.06	2.12	2.12	2.84
1250								
1300			1103	306				
1500	2811	1120	336					
1600				1608	2.24	2.45	2.41	2.15
1700			491					
1800		1048						
2000	2872		513					
2075		998		1490	2.38	2.75	2.68	2.28
2300			589					
2400	2126	951	632	1478	2.34	2.83	2.77	2.35
2850		941 ⁺						
3000	2180		724					
3090				1564	2.16	2.61	2.63	2.53
3150	2297							
3300	2284		769					
3500	2388							
3508			802	1696	2.00	2.79	2.95	2.93
3700	2497							
3800			828					
4080	2696		841	1813	2.04	2.58	3.07	3.22
4200	2742							
4400	2760 ⁺							
4600			844 ⁺	1861 ⁺	2.10 ⁺	2.35 ⁺	3.14 ⁺	3.34 ⁺
6200		941 ⁺		1861 ⁺	2.10 ⁺	2.35 ⁺	3.14 ⁺	3.34 ⁺
6600		960						
9000	2765 ⁺	1821	844 ⁺	1866	2.19	2.52	3.31	3.69
9200		1897	822					
9750	2748		773	1902	2.21	2.19	3.01	3.29
10000		1242						
10300			686					
10800	2694	1441	604	2935	2.25	2.05	2.52	2.72
11300			514					
11850	2613	1730	415	2146	2.27	2.66	2.18	2.33
12300			323					
12050	2532	1851	234	2294				
13000			151					
13570	2306 ⁺	1930 ⁺	35 ⁺	2279	2.08 ⁺	2.08 ⁺	2.08 ⁺	2.09 ⁺

mm (exponents dimensionless) · * fixed · = tied (constant cabin section)
pchip (monotone cubic) through the knots of each line

SECTION LAW (each half)

$$|y / h_w|^n + |(z - z_{mw}) / h_z|^m = 1$$

lw = half-breadth, zmw = max-breadth WL,
hz = crown - zmw (upper) or zmw - keel (lower);
n: crown / keel flatness, m: side flatness;
n = m = 2: ellipse. Section parameter t = polar
angle about (0, zmw) in normalised coordinates.

DEVIATIONS OF THE LOFT FROM THE PILATUS DRAWING (NORMAL DISTANCE, mm)

DATASET	N	RMS	MAX	REV A RMS	REV A MAX	WORST AT	REMARK
PROFILE CROWN	794	1.0	12.9	109.5	141.1	STA 1060	max at the cowl front: drawn nose 18 high (drawn prop axis WL 1673; WL 1655 kept)
PROFILE KEEL	749	2.1	11.7	178.2	333.9	STA 1060	visible keel only (not behind wing fairing 5300-7400, strake 10720-11960); max as crown
HALF-BREATH	1164	0.9	3.1	90.8	230.7	STA 12740	view keel only (not behind wing fairing 5300-7400)
FIN-ROOT LINE	161	1.6	3.4	118.1	137.5	STA 12450	dorsal fin root line (side view) at the plan lift half-breath. STA 9250-12450
PNR001 SHOUJETTE	296	0.0	0.0	0.0	343.3		lookage never below the strake / ventral fairing silhouette (STA 10700-13560). MAX = protrusion
EF1 1200	68	4.0	14.5	68.0	95.8	BL 254 WL 1559	cowling ring only; the chin-inlet crescent is not OML (Stage 3 inlet)
EF2 2075	133	0.7	1.5	72.2	134.7	BL 292 WL 1042	
FR10 3080	133	1.9	3.8	28.7	54.2	BL 135 WL 2182	
FR12 3090	133	2.8	5.9	34.7	93.1	BL 731 WL 2096	
FR14 4080	144	3.2	6.9	55.7	130.5	BL 338 WL 961	
FR16-30 4590-8230	173	2.3	8.2	77.8	139.1	BL 337 WL 965	constant cabin section, checked at 5 stations 4590-8230
FR33 9000	223	4.3	11.4	90.2	212.1	BL 818 WL 1444	keel knee region (flat bottom to tail-cone transition)
FR36 9750	129	1.6	3.4	116.6	295.9	BL 745 WL 1550	dorsal fin excluded
FR38 10860	89	0.9	2.5	244.3	314.3	BL 0 WL 2690	dorsal fin, ventral strakes excluded
FR40 11850	116	0.5	1.2	207.1	350.4	BL 602 WL 2763	dorsal fin, strakes, ventral fairing (BL 121, WL 1600) excluded
FR19-FR31 (SHEET 2)	840	4.3	12.1	66.0	139.8	FR31 BL 0 WL 939	phantom cabin section, weight 0.25 (sheet 2, x r-15, z r-45). FR31 lies on the keel knee

Targets: profile / half-breadth RMS ≤ 8 , max ≤ 20 ; every frame section max ≤ 20 - ALL MET. REV A = the model before this refit

CABIN SECTION NUMBERS

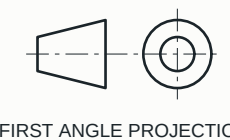
CABIN (STAGE 3)	mm
crown / keel WL (STA 4600-8200)	2769 / 941
OIML depth + max breadth	1828 / 1588
max-breadth WL	1801
ceiling WL (crown - 40)	2729
floor WL (ceiling - 1470)	1259
floor WL change vs rev A (1120)	+139
OIML width at the floor	1545
floor width avail. (c 2 x 85)	1375 ~ 1300
max inner width (c 2 x 85)	1518 ~ 1520

NOTES

1. Drawn from the left parameters of modelisefuella (key/ knot labels + section label, not from the mesh). Spinnars = STA m Att of the datum (3000 ahead of the firewell), WL above the static ground line; BL from the centre line - starboard.
2. Knot values fitted by least squares (drawings/lines, Rly) to the registered Platina NOX model drawing 190.10.40.032; side crown / visible keel, plan half-Broad frame sections EFJ-FR40 (sheet 1) and the sheet 2 phantom cable with smoothness regularisation. Deviation = normal distance, + = outside.
3. Fixed, spinner lift STA 300, prop axis WL, 1655 at the disc (prop clearance 320 keel; the drawing shows 320; thus side crown / visible keel 2 / dge right / frame), so the spinner is at WL 1636 and the spinner base centre 4 mm above the cow-lift ring (R 250 at STA 1094, Stage 3; firewell STA 3000, tail cone closure STA 13570 at the lower end of the rudder ladder length; 14,000 overall length is set by the tail bulb).
4. Not part of this OM, (Stage 3 parts): chin-ribs/cross dorsal, fin, ventral strakes, ventral leading under the tail cone (FR40, lower shooteout of STA 11900), wing-rafter. Aft of the rudder leading edge (STA 12650) at the keel) the tail cone lies inside the rudder / is trimmed there.
5. Changes vs rev A: cabin crown 3200 > 2769, keel nose 800 > 941, (flat from STA 1094), aft body slimmer (half-Broad frame STA 11000 740 > 569), lat end 14360 > 13570, spinner R 230 > 250, cowt front 950 > 1040.

PILATUS PC-12 PRO

FUSELAGE LINES PLAN



DWG NO	REV	SCALE	SHEET	UNITS
PC12-L1	A	AS SHOWN	A1 · 1 OF 1	mm
DRAWN				DATE
Claude - loftkit parametric model				2026-09-24

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION